



NITI Aayog

Strategy for

New India@75

STRATEGY FOR NEW INDIA @ 75

MODULE - 18

18

DIGITAL CONNECTIVITY

OBJECTIVES



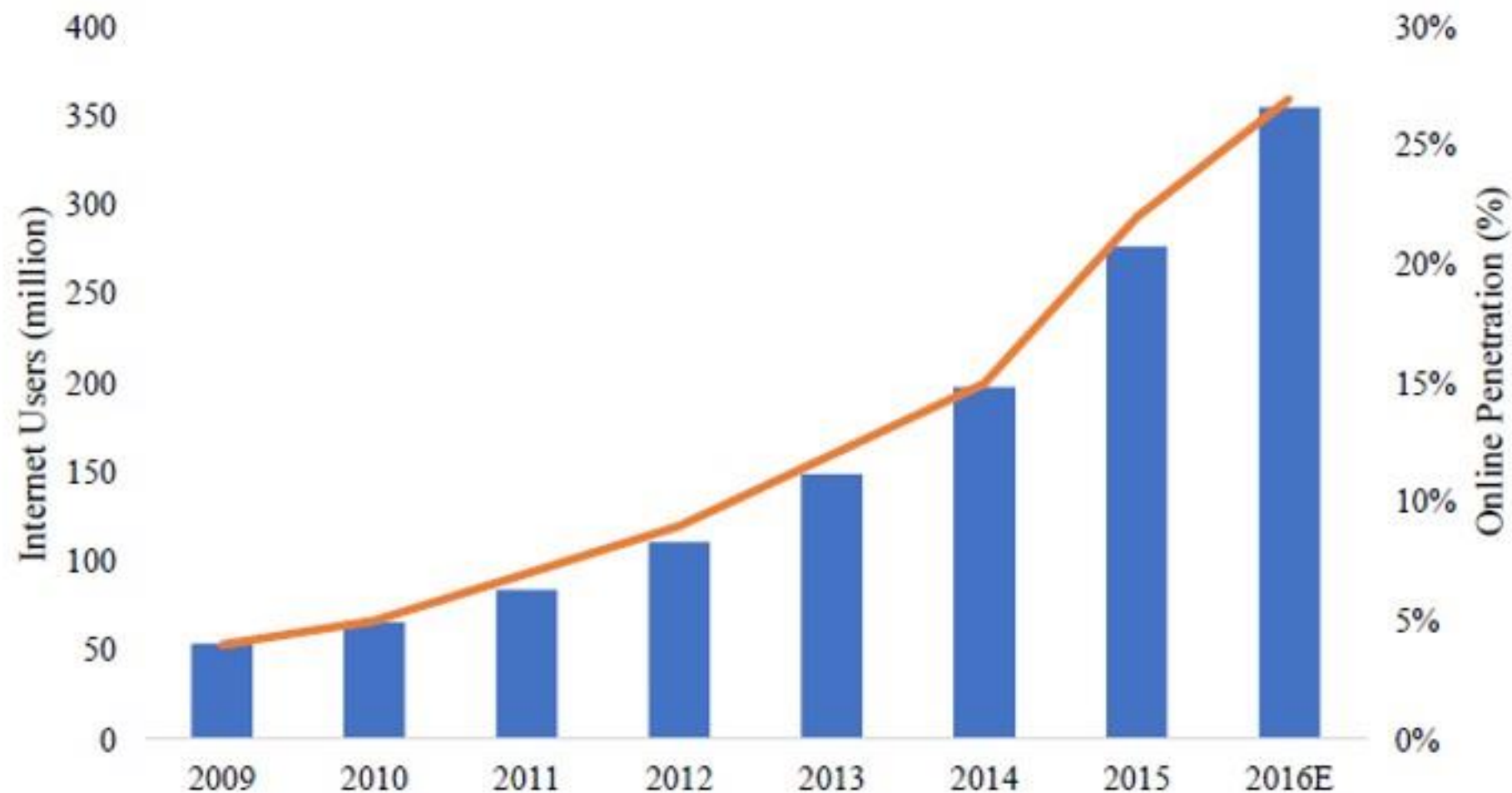
- Relevance of digital connectivity to economic growth is well known.
- The need is to **eliminate the digital divide by 2022-23**.
- It requires physical digital connectivity across all the States, Districts and Gram Panchayats.
- Delivery of Government services digitally by 2022-23.
- **100% basic digital literacy** across the Country to enable the benefits of digitization.



नीति आयोग

Federal Institution for Transforming India

Figure 18.1: Increase in internet users and online penetration in India



Source: Internet Trends 2017 by KPCB³



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DIGITAL CONNECTIVITY

PRESENT SCENARIO



- As per **Internet Trends 2017 Report**, 355 million people i.e., **27% of India's population** uses the internet. It was only just 4% in 2009.
- Under **Digital India Scheme**, construction of broadband highways, public internet access, e-governance and development of basic information technology skills achieved considerable progress.
- In 2011, Scheme for the creation of a **National Optical Fibre Network (NOFN)** was initiated to connect all the Gram Panchayats (GPs) of the Country with high-speed internet.
- Though it is lagging behind, but substantial progress has been made.

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DIGITAL CONNECTIVITY

PRESENT SCENARIO



WHAT ABOUT OTHER EFFORTS?

- The **National Information Infrastructure (NII)** will ensure the integration of the networks and cloud infrastructure to provide high-speed connectivity to various Government departments up to the panchayat level.
- The **components of NII** include ---
 - State Wide Area Network (SWAN)
 - National Knowledge Network (NKN)
 - BharatNet
 - Government User Network (GUN)
 - MeghRaj Cloud

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DIGITAL CONNECTIVITY

PRESENT SCENARIO



OTHER EFFORTS OF THE GOVERNMENT

- The Government also launched **Public Internet Access Programme** to make **2,50,000 Common Service Centres operational** at the gram panchayat level to deliver Government services online.
- Under this programme, 1,50,000 post offices will be converted into multi-service centres.
- The last mile connectivity, through Wi-Fi or any other suitable broadband technology, is to be provided at all GPs in the Country, funded by the **Universal Service Obligation Fund (USOF)**.
- A comprehensive plan to cover remote villages in the North-East, which does not have mobile coverage, are identified and action has been initiated in phases.

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DIGITAL CONNECTIVITY

CONSTRAINTS

1 BROADBAND CONNECTIVITY

- Internet access is plagued by issues related to **quality and reliability**, **outages**, **call drops** and **weak signals**.
- The current definition of broadband of **512 kbps speed** is **inadequate** and is not in line with the expected rise in demand in the future.
- Existing networks are strained by **limited spectrum availability and usage**, affecting the provision of quality services.

2 DIGITAL ACCESS AND LITERACY

- A significant portion of our population does not have access to devices such as laptops, computers, smartphones, etc.
- Digital literacy in India is estimated to be less than 10% of the population.



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DIGITAL CONNECTIVITY

CONSTRAINTS

3 CONTENT IN INDIAN LANGUAGES

- Currently, most digital content is in English.
- However, KPMG Report suggested that “9 out of every 10 new internet users in India over the next 5 years are likely to be Indian language users”.



4 AVAILABILITY OF E-SERVICES

- A large number of e-services are not available on the digital platform.
- There is wide variation across states in the availability of citizen e-services.
- Currently, citizens **have to physically visit Government Offices to access most Government-to-Citizen (G2C) services**, as Municipalities and other Government Bodies are slow to digitize their processes.

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DIGITAL CONNECTIVITY

CONSTRAINTS



5

CYBER SECURITY

- The regulatory framework for cyber security is inadequate.
- **Hacking and denial-of-service attacks** have led to disruption of services, both in the Government and the Private Sector.
- Banks and Governments increasingly face security breaches.

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DIGITAL CONNECTIVITY

WAY FORWARD

1

BROADBAND CONNECTIVITY



Ministry of Electronics and
Information Technology
Government of India

- A modified strategy for expanding the pace of implementation of **BharatNet** was approved and as part of the strategy, remaining 1.50,000 Gram Panchayats are to be connected in Phase-II through state-led model, private sector model and CPSU model by using an optimal mix of media (OFC, radio and satellite).
- **Service delivery mechanism**, by providing **last mile connectivity through Wi-Fi** or any other suitable broadband technology to all GPs, has been made an integral part of the project.
- **Public Wi-Fi hotspots** are being set up by BSNL at its 25,000 telephone exchanges in rural areas and Wi-Fi choupals are being set up in 5,000 GPs by CSC-SPV under the Ministry of Electronics and Information Technology.

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DIGITAL CONNECTIVITY

WAY FORWARD

1

BROADBAND CONNECTIVITY ... *Contd*



- Provision of universal broadband coverage at 50 Mbps to every citizen and enabling of 100 Mbps broadband on demand to all key development institutions, including educational institutions, are part of the new National Digital Communications Policy, 2018.
- TRAI should consider putting in place a credible system to track call drops, weak signals and outages to ensure the quality and reliability of telecom services.

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DIGITAL CONNECTIVITY

WAY FORWARD

2

QUALITY OF SERVICE



- Adequate spectrum availability is critical to ensure service quality.
- Efficient spectrum allocation in large contiguous blocks should be explored.
- We should also explore migration to new technologies.

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DIGITAL CONNECTIVITY

WAY FORWARD

3

ACCESS AND DIGITAL LITERACY



- Digital literacy needs special focus at the school / college levels.
- The **National Digital Literacy Mission** should focus on introducing digital literacy at the primary school level in all Government schools for basic content and in higher classes and colleges for advanced content.
- It will have **multiplier effect**, when these students in turn educate their family members.
- Higher digital literacy will also increase the adoption of computer hardware across the Country.

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DIGITAL CONNECTIVITY

WAY FORWARD

4

CONTENT IN INDIAN LANGUAGES



- State Governments should pay special attention in creating content, particularly those relating to government e-services, in Indian regional languages.
- To fulfil the vision of making **all Government's online services available in all 22 Official Languages**, there needs to be focused collaboration between the Centre, States and Researchers to promote Natural Language Processing (NLP) in Indian languages.
- Focus could be laid on the **automatic translation of content into regional languages** from Hindi or English.
- **Case studies like those of the European Union**, which had similar problems, should be explored to identify the models that can potentially be adopted.

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DIGITAL CONNECTIVITY

WAY FORWARD

5

AVAILABILITY OF E-SERVICES



- Individual Ministries and States have to play a pivotal role in ensuring that all their services are available and easily accessible by citizens over digital platforms.
- **Digital platforms**, that enable real-time data updates, should **increase accountability, and facilitate monitoring, quality checks and timely intervention** by the higher administrative authorities.
- Procurement of such digital platforms could be standardized by the Central Government based on an analysis of successful case studies in India.

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DIGITAL CONNECTIVITY

WAY FORWARD

6

CYBER SECURITY



Ministry of Electronics and
Information Technology
Government of India

- Ministry of Electronics and Information Technology (MeitY) will need to evolve a **comprehensive cyber security framework** for data security, safe digital transactions and complaint redressal.
- The **National e-governance Division** of MeitY should periodically audit compliance of e-services offered by State Governments.
- It should bring out a performance report of e-services with a view to improve service delivery.

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